

The Metropolis Method on the Ising Model and Ferromagnetism

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The study of ferromagnetism within the framework of the Ising Model provides insights into the behavior of magnetic systems. This research utilizes the Metropolis algorithm to simulate the Ising Model on various lattice sizes, aiming to understand the effects of temperature on the system's evolution and the nature of phase transitions. By examining the magnetization per spin and calculating specific heat and magnetic susceptibility, the work investigates the difference in behavior at varying temperatures. The findings contribute to a deeper understanding of first and second-order phase transitions, revealing the correlation between temperature and spin behavior in ferromagnetic materials.

I. INTRODUCTION

The Ising Model is a basic model of ferromagnetism depicted by a periodic lattice of "spins". These spins are related to an electron's magnetic moment. A ferromagnetic collection of spins is characterized by having a majority of spins that point in the same direction. Such collection of spins will also have a lower energy than a collection that had more spins in disagreeing directions. The existence of higher energy states occur due to temperature and thermal equilibrium. This is relevant in physics because the Ising Model is a basic way of studying quantum mechanics and statistical mechanics. It also provides insight on first and second-order phase transitions.

The system's behavior is governed by the energy equation below, where J represents the strength of interaction between spins, and H represents the strength of interaction with a magnetic field. s can either be $+1$ or -1 and represents the orientation of a specified spin.

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - H \sum_i s_i \quad (1)$$

This system, as with all systems, will evolve towards states with lower energies. However, this is affected by the surrounding temperature, such that the probability, P_α of a system being in any given microstate α is given by the similarity below, where E_α represents the energy of the microstate, k_B represents Boltzmann's constant, and T represents the surrounding temperature.

$$P_\alpha \sim e^{-\frac{E_\alpha}{k_B T}} \quad (2)$$

The objectives of this project are to see the effect of temperature on the evolution of ferromagnetic systems as well as to see first and second order phase transitions in the Ising Model.

II. METHODS

A. Normalization Process

We will be using normalized values $\bar{E}$, $\bar{T}$, and $\bar{H}$ such that $\bar{E} = E/E_0$, $\bar{T} = T/T_0$, and $\bar{H} = H/H_0$ where $E_0 = H_0 = J$ and $T_0 = E_0/k_B$. This is equivalent to setting $|J|$ and k_B equal to 1. This transforms Equation 1 into the normalized version below.

$$\bar{E} = - \sum_{\langle i,j \rangle} s_i s_j - \bar{H} \sum_i s_i \quad (3)$$

From now on, the bars will be dropped on these terms, and they should be understood to be the normalized terms unless otherwise stated. When comparing Equation 3 to Equation 1, it is clear how the normalized version is identical to setting $J = 1$. Similarly, the normalized version of Equation 2 becomes $e^{E_\alpha/T}$, which is equivalent to setting $k_B = 0$.

B. Simulation Approach

The lattice is represented by an array of spins valued at $+1$ or -1 . Conditions are simulated with a Monte Carlo method called the Metropolis algorithm. This involves sweeping across every spin and flipping the sign. If this causes the microstate to lower in energy, then the spin stays flipped. If the microstate increases in energy, then a random number is picked from 0 to 1 and compared to the Boltzmann factor $e^{\Delta E/T}$. The spin stays flipped only if the random number is less than the Boltzmann factor. When the entire array of spins have been swept through, this is considered a single Metropolis step.

We will be analyzing a first-order phase transition through a 128×128 sized lattice, initializing with all spins up. For a temperature of $T = 1$, we sweep through values of H starting from -5 to 5 and then back down to -5 . For each change of H , we allow 1000 Metropolis Steps to run. We collect the ensemble magnetization per spin for every H value. This is repeated for a temperature of $T = 4$. The ensemble magnetization per spin is given by the equation below, where N_{MC} represents the number

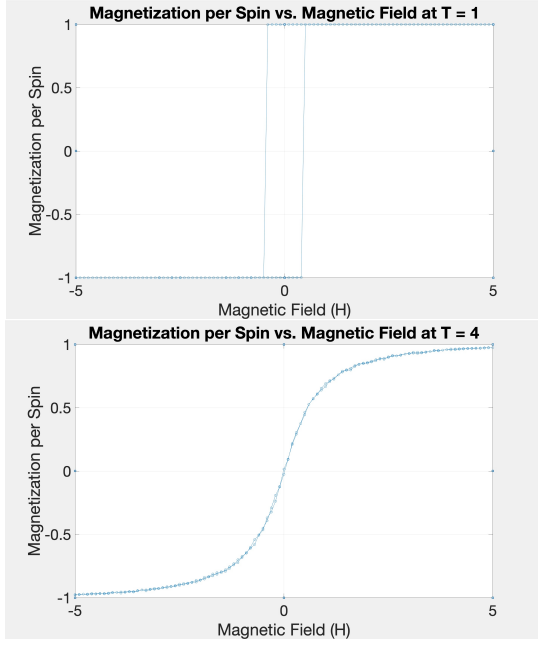


FIG. 1: Magnetization per Spin vs. Magnetic Field at $T = 1$ (top) and $T = 4$ (bottom).

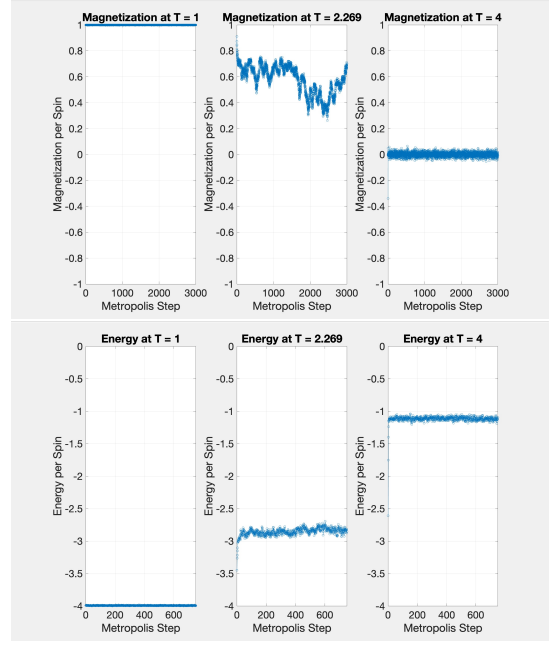


FIG. 2: Magnetization (top) and Energy (bottom) per Metropolis step for varying temperatures.

of microstates created by the Monte Carlo simulation.

$$\langle \bar{M} \rangle = \frac{1}{N_{MC}} \sum_{\alpha=1}^{N_{MC}} \bar{M}_{\alpha} \quad (4)$$

To study second-order phase transitions, we will be collecting the magnetization per spin at every Metropolis for three different temperatures. This will be used to determine how many Metropolis steps are required to reach thermal equilibrium. We then calculate the value of two thermodynamic quantities, specific heat and magnetic susceptibility. We aim to see how these quantities change depending on temperature for different lattice sizes. The Equations 6 and 6 below were used to calculate specific heat and magnetic susceptibility, respectively.

$$C_V = \frac{1}{N_s} \frac{1}{k_B T^2} (\langle E^2 \rangle - \langle E \rangle^2) \quad (5)$$

$$\chi = \frac{1}{N_s} \frac{1}{k_B T} (\langle M^2 \rangle - \langle M \rangle^2) \quad (6)$$

N_s represents the total amount of spins in the system.

III. RESULTS

The results of the study of first-order phase transitions can be seen in Figure 1, which are plots of magnetization per spin versus magnetic field (H) for temperature values of 1 and 4.

The study of second-order phase transitions first include a justification of using only 200 metropolis steps per temperature. This is depicted in Figure 2 as a series

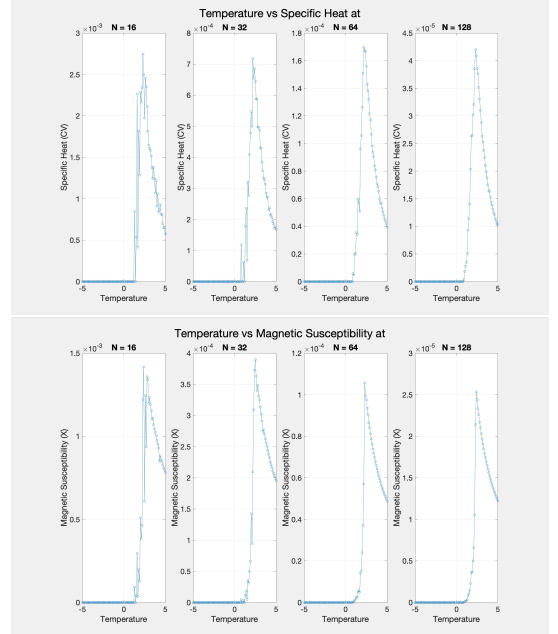


FIG. 3: Specific Heat (top) and Magnetic Susceptibility (bottom) vs. Temperature for varying lattice sizes.

of plots indicating the magnetization and energy quantities per metropolis step at varying temperatures. Further justification is provided in the Discussion section below.

The second part of this study is an examination of the specific heat and magnetic susceptibility versus temperature for varying lattice sizes, depicted in Figure 3.

IV. DISCUSSION

The top of Figure 1 shows properties of a first-order phase transition. This can be seen from the existence of discontinuities in the graph. Another important property at this temperature is the existence of a hysteresis loop. Hysteresis is described as the dependence on the previous state. This can be seen at the top of Figure 1 from the rectangular loop located at the center of the plot. If the magnetization per spin is previously negative, then the magnetic field must reach a value slightly greater than 0 in order to cause all the spins to flip simultaneously; however, if the magnetization per spin is previously positive, then the magnetic field must reach a value slightly less than 0 in order to cause all the spins to flip simultaneously. This can be contrasted by the bottom of Figure 1, which shows properties of a second-order phase transition. This can be seen by the continuity of the graph. The discontinuity will only show in the plot of it's derivative, which would be located at $H = 0$. It is classified as a second-order phase transition due to this fact. Additionally, the hysteresis loop disappears in these conditions.

The only difference between the two conditions is that the bottom plot is simulated at a higher temperature. This suggests the existence of a critical temperature where the behavior can be expected to go from showing a first-order phase transition to a second-order phase transition. Physically, this can be justified with statistical mechanics. At very low temperatures, it is highly improbable for non-aligned states to exist due to them being higher energy states. Increasing the temperature makes these higher energy states more probable, which allows for a smoother transition between all spins down to all spins up when increasing the magnetic field.

Figure 2 shows extremely convergent behavior for temperatures far away from $T = 2.69$, the critical temperature. It takes less than 20 Metropolis steps at $T = 1$ and $T = 4$ in order to converge to values for magnetization and energy. In contrast, these values do not seem to every converge well after more than 3000 metropolis steps at $T = 2.269$. To obtain the graphs in Figure 3, we choose to do 200 metropolis steps per temperature value. This is a compromise between the rapidly converging behavior for $T = 1$ and $T = 4$, and the erratic behavior for T at the critical temperature. 200 was chosen because

the plot for critical temperature in Figure 2 seemed to finish exhibiting the initial jump in value after around 200 steps.

Additionally, Figure 2 shows the difference in end behavior for temperatures below and above the critical temperature. For temperatures below the critical temperature, we see the values converge to fully magnetized ($M = 1$) and existing at the lowest possible energy state. In contrast, we see 0 magnetization for higher temperatures and the energy converging to a higher value.

From Figure 3, we expect these values to diverge to infinity at the critical temperature for infinitely large lattices. The peaks in each graph can be found around $T = 2.7$, which is close to the critical temperature. It is very unexpected to see the peaks decrease in value as the lattice size increases because we should expect these peaks to approach infinity as the the size approaches infinity. This may be caused by the normalization process, where the normalization keeps the value contained within a finite bound. Nevertheless, Figures 2 and 3 still depict a shift in behavior at the critical temperature and non-convergent behavior exactly at the critical temperature.

V. CONCLUSION

This investigation into the Ising Model via the Metropolis algorithm reveals several key aspects of ferromagnetic behavior. The occurrence of first-order phase transitions is characterized by the presence of discontinuities and hysteresis loops at low temperatures, emphasizing the system's history-dependent properties. Conversely, second-order phase transitions are continuous along the whole function and only discontinuous at the first derivative. The critical temperature exists at the point where the system goes from favoring lowest-energy aligned states to higher energy, misaligned states. The computational results for specific heat and magnetic susceptibility align with theoretical predictions, depicting pronounced peaks near the critical temperature. Although the decreasing in peak value with increasing lattice size deviates from expectations, it highlights the influence of finite-size effects and normalization processes on computational models. Further research could include properties of a three dimensional system of spins or the effects of a non-uniform magnetic field. Another field of study can be investigated regarding the existence of higher-order phase transitions.

[1] N. Giordano and H. Nakanishi, *Computational Physics* (Pearson Prentice Hall, Upper Saddle River NJ, 2006).